

AutoMorph

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Chapter 1

Introduction

This document is an overview for our project "AutoMorph", which aims to improve the performance of driver safety systems by extending existing datasets. We will improve the current data by generating synthetic images of drivers in underrepresented scenarios, and this will be done using diffusion models.

1.1 Problem Domain

Driver Safety is an important concern for the trucking sector, more so in developed countries with their vast long-haul operations cross-state. Due to these long-haul operations, truck drivers often spend long hours on the road, increasing the chances of tiredness and thus, the likelihood of dangerous driving. Ensuring the safety of these truck drivers indirectly ensures the safety of the road network as well as the cargo they are transporting.

So far, various driver safety systems have been implemented to monitor these truckers on their journey. These systems are specifically designed to detect high-risk behaviors such as yawning, eating, using their mobile while driving, etc. By monitoring these actions, alerts can be given to remind the driver about their behavior prevent potential accidents, and ensure no loss to the transportation company. Henceforth, minimize the risk to both the driver and other road users, contributing to the overall safety of transportation networks.

However, these driver safety systems are not the golden standard yet, they're only as good as the data they are trained on and thus, face a ton of issues when handling diverse or underrepresented behaviors because it is simply too difficult to acquire training data to fulfill these two criteria.

1.2 Research Problem Statement

Current AI-based driver monitoring systems, such as those developed by Motive and other industry leaders, focus on detecting high-risk behaviors such as yawning, eyes closed, eating, drinking, mobile phone usage, seat belt violation, etc. These behaviors are often one of the leading reasons why fatal crashes happen in truck driving. While AI-based driver monitoring systems have evolved considerably in the past few years, their performance is still majorly defined by the data upon which they are trained.

These models are trained using large amounts of annotated real-life data to effectively learn to predict high-risk behaviors. However, some of these behaviours such as fatigue-induced yawning or not wearing a seatbelt, are often underrepresented in various datasets. For example, suppose there is a 24-hour video of a person driving, around less than 1 percent of that timeframe would include a person yawning. So when there is a lesser quantity of this type of data, the features predicting this behavior would have a lower precision and recall. This lack of diversity and well-distribution of the dataset will lead to suboptimal performance. Consequently, the system may not detect a fatigued driver, potentially resulting in life-threatening accidents.

Moreover, the process of collecting and annotating driver behavior is very costly and time-consuming. Without a more efficient way for this procedure, the effectiveness of this system will remain limited.

1.3 Software Requirements Specification

This section outlines the functional and non-functional requirements for AutoMorph, which improves driver monitoring systems by generating synthetic images through diffusion models. The software will support image augmentation techniques such as inpainting, pose augmentation, and text-conditioned generation to create realistic synthetic data.

The successful implementation of AutoMorph depends on meeting the following criteria:

1.3.1 Functional Requirements

1. Driver Image Augmentation:

- The system shall augment existing driver-facing images with underrepresented risky behaviors such as yawning, and eyes-closing.
- Inpainting techniques will be used to target specific regions of the image for augmentation (e.g., mouth for yawning, eyes for closure).

2. Pose Augmentation:

- The system shall modify the posture of the driver using a DragDiffusion-like functionality. (e.g., shifting the position of a phone or turning the head).

3. Text-Based Conditioning:

- The system shall support text-based prompts to guide the image generation process, enabling controlled outputs based on the specific conditions described in the text (e.g., “Generate a person yawning with eyes closed”).

4. Dataset Integration:

- The augmented images shall be seamlessly integrated into existing datasets to improve the accuracy of AI-based driver monitoring systems.

1.3.2 Non-Functional Requirements**1. Performance:**

- The synthetic image generation process should operate efficiently, augmenting at least 20-30% of the existing dataset within a reasonable timeframe, without compromising quality.
- Generated images must closely mimic the real-world counterparts, ensuring that the synthetic data does not appear artificial to AI systems during training.

2. Scalability:

- The software should support scalable generation, handling an increasing number of images or new classes of driver behaviors without significant degradation in performance.

3. Usability:

- The system shall provide an interface for easy integration of augmented images into the existing driver monitoring AI pipeline for retraining and performance evaluation.

Chapter 2

Literature Review

Before choosing a valid approach, we went over several literature encompassing various diffusion model related techniques. We will summarize each paper under 3 subsections: Summary, Analysis, Use in Project.

2.1 Related Research

2.1.1 High-Resolution Image Synthesis with Latent Diffusion Models (2022)

2.1.1.1 Summary of the research item

This article presents Latent Diffusion Models (LDMs) for the task of high-resolution image synthesis. LDMs extend the diffusion process over a latent space. Thereafter, this latent is rendered into a high-res image. This approach cuts computation cost by a large degree compared to the pixel-space diffusion models and at the same level or even better image quality is obtained.

2.1.1.2 Critical analysis of the research item

They argue that the efficient and scalable approach of this paper for high-resolution image synthesis is quite novel. However, that comes with difficulties as the latent space representations employed are rather critical and may be problematic with large dimensional datasets or detailed images. The extent to which this will affect practical applications remains to be established.

2.1.1.3 Relationship to the proposed research work

This paper will form the basis for understanding latent diffusion models and in particular their application in high-resolution image synthesis.

2.1.2 Denoising Diffusion Implicit Models (DDIM) (2021)

2.1.2.1 Summary of the research item

DDIM uses a new framework for denoising diffusion models that have improved sampling efficiency. They utilize implicit models that allow faster generation of samples whilst maintaining relatively high-quality outputs. Their approach in particular has reduced the number of steps required for the diffusion process which in turn, accelerates the image synthesis process.

2.1.2.2 Critical analysis of the research item

DDIM provides major improvements with respect to sampling speed. Their method may introduce fidelity issues which is a fair trade-off considering the speed boost they have achieved.

2.1.2.3 Relationship to the proposed research work

DDIM will be looked at and used in the image generation process to speed up the image synthesis process.

2.1.3 DragDiffusion: Harnessing Diffusion Models for Interactive Point-based Image Editing (2023)

2.1.3.1 Summary of the research item

DragDiffusion is a successor paper to DragGAN where an image's context is morphed by altering the spatial coordinates of the image's latent representation during the first few denoising steps.

2.1.3.2 Critical analysis of the research item

DragDiffusion allows a modest improvement in image modification however, this method does struggle in semantically rich edits.

2.1.3.3 Relationship to the proposed research work

DragDiffusion's method will be recreated and used for pose augmentation during Iteration number 3, allowing us to modify the pose of the driver under different contexts.

2.1.4 SDXL: Scaling Diffusion Models for High-Resolution Image Synthesis (2023)

2.1.4.1 Summary of the research item

Stable Diffusion XL focussed on scaling their predecessor model (Stable diffusion) for an even better performance on high quality image synthesis. The paper delved into various techniques for optimizing model architecture as well as introducing its own VAE and a two-stage process with the second stage refining the image.

2.1.4.2 Critical analysis of the research item

Albeit this model has significantly improved diffusion model, the scalability of this model is very limited due to the sheer amounts of VRAM the model requires.

2.1.4.3 Relationship to the proposed research work

If there are no hardware blockers on utilizing this architecture then we will upgrade our backbone latent diffusion model to this one. In worst-case scenario, the conditioning techniques can be used from this paper.

2.1.5 RePaint: Inpainting using DDPM (2022)

2.1.5.1 Summary of the research item

RePaint introduces an inpainting method using denoising diffusion probabilistic models. They focus on inpainting parts of the image through the diffusion process, whilst maintaining detailed and coherent results in the process.

2.1.5.2 Critical analysis of the research item

RePaint opens the doors to inpainting in a detailed matter however it is still shrouded by inpainting complex semantic meaning such as occluded objects etc.

2.1.5.3 Relationship to the proposed research work

Perhaps the most important part of the project, the inpainting process from this paper will be used to augment existing driver faces and generate new synthetic results.

2.2 Tabular Analysis Summary of Research Items

Paper Title	Method	Dataset	Research Gap
High-Resolution Image Synthesis with Latent Diffusion Models (2022) [1]	A VAE compresses the entire image space into a smaller latent space. The latent space is used for training a diffusion model (U-Net) that deals with 2D noise prediction inside the latent representation. Afterward, remove noise from a randomly sampled noisy latent which is then decoded into the final predicted image.	CelebA-HQ, LSUN	LDMs have a slower generation speed than that of GANs. Further improvements can be made in terms of text/spatial conditioning.
Denoising Diffusion Implicit Models (DDIM) (2021) [2]	A deterministic formulation of the diffusion process was created, with just one last step being non-deterministically defined. This enables sampling without the need to go through retraining and also considerably accelerates the process of generation.	CIFAR-10, CelebA-HQ	The DDIM models, while trying to speed up the sampling time cause these new sampling methods to generate lower quality images in comparison to what can be achieved using stochastic sampling methods.
DragDiffusion: Harnessing Diffusion Models for Interactive Point-based Image Editing (2023) [3]	The method presents a 3-stage process: LoRA-based identity preserving fine tuning, user driven latent optimization (motion supervision and point tracking), and denoising with reference latents.	DragBench	The paper still leaves room for future research in further improving the responsiveness of interactive editing systems and scaling the technique for more complex semantic image edits

Table 2.1: Comparison of diffusion-based model approaches and inpainting techniques in recent literature

Paper Title	Method	Dataset	Research Gap
SDXL: Scaling Diffusion Models for High-Resolution Image Synthesis (2023) [4]	The SDXL (Stable Diffusion XL) is an improvement of the original stable diffusion where more parameters were added for controlling text-image correspondence and finer conditioning and other additional features were provided like size and crop conditioning. Generation of images of multiple aspect ratios has also been introduced as well as a better VAE which improves the clarity of images. In addition, there is a refining stage for SDXL to improve backgrounds and details.	Custom SDXL datasets	SDXL employs two models which affects accessibility and sampling speed due to the need to load two large models. Maybe we can combine these two models into one large one.
RePaint: Inpainting using Denoising Diffusion Probabilistic Models (2022) [5]	The paper proposes an algorithm for inpainting the areas that are known and areas that are to be filled in, and then, inpainting is done to the content whilst maintaining consistency with known areas of the image over repeated passes.	CelebA, FFHQ, ImageNet	Complex inpainting situations such as Image occlusion still remain hard to inpaint properly.

Table 2.2: Comparison of diffusion-based model approaches and inpainting techniques in recent literature

Chapter 3

Proposed Approach

3.1 Dataset

For this project, we will be utilizing the **100-Driver dataset**, which consists of 470,000 images taken from 4 cameras that observe 100 drivers over 79 hours of driving from 5 different vehicles. This dataset offers substantial variability, which makes it ideal for building robust driver safety systems. The 100-Driver dataset covers key variations that are crucial for real-world applications, including:

- Changes in vehicle types
- Variations in driver appearances and behaviors
- Different camera viewpoints
- Diverse lighting conditions (day/night, shadows, etc.)

The presence of these variations makes the dataset highly representative of real-world scenarios, thus ensuring that our synthetic data augmentations are both effective and practical.

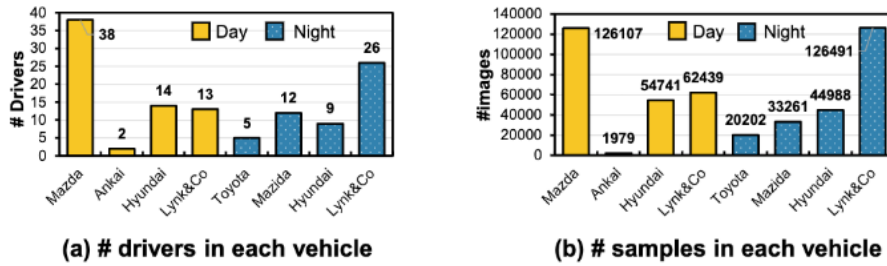


Figure 3.1: Samples Distribution based on the type of car.

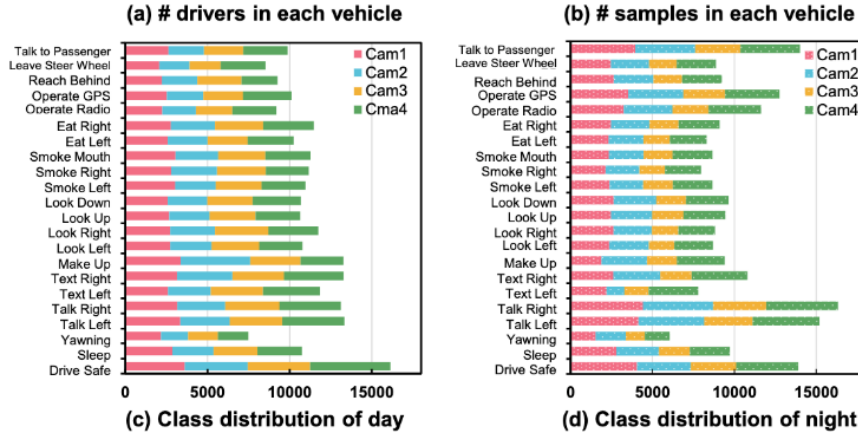


Figure 3.2: Class Distribution of images for daytime and nighttime.

3.1.1 Dataset Preprocessing

Given the vast size of the dataset, preprocessing is required to handle the large-scale image data effectively. Images will be resized to a consistent dimension, and irrelevant parts of the images will be filtered out based on the bounding box data obtained from the next step.

3.2 Bounding Box Extraction with Owl-ViT

To ensure that we focus on key regions of interest, such as the driver's face, hands, and objects in the environment, we will employ **Owl-ViT v2**. Owl-ViT (Object-aware Vision Transformer) allows us to perform image-caption-based detection. By providing an image along with a prompt describing the object or region of interest, Owl-ViT will output bounding boxes with associated confidence scores.

3.2.1 Process

The process for extracting bounding boxes from images in the 100-Driver dataset using Owl-ViT consists of the following steps:

1. Input an image from the dataset into Owl-ViT v2.
2. Provide a prompt mentioning the object or area of interest (e.g., "driver's face", "steering wheel", "mobile phone").
3. Owl-ViT processes the image and outputs a bounding box around the specified area along with a confidence score.

4. Store the image file path and the best-predicted bounding box in a dictionary-like structure for future use in synthetic data generation.



Figure 3.3: Example of bounding box extraction using Owl-ViT on a driver image.

3.2.2 Dictionary Structure

The bounding boxes and image file paths are stored in a dictionary for easy access and later augmentation. An example structure would look like this:

```
{
  "image_path_1.jpg": {
    "bbox": [x_min, y_min, x_max, y_max]
  },
  "image_path_2.jpg": {
    "bbox": [x_min, y_min, x_max, y_max]
  }
}
```

This data structure allows efficient referencing and retrieval of bounding boxes during the image augmentation process using diffusion models.

3.3 Dreambooth Inpainting for Yawning Model

In this section, we propose a method to train a yawning model by leveraging a Dreambooth-like inpainting approach. The goal is to generate synthetic images of drivers yawning by manipulating existing images through a combination of latent space encoding, masking, and noise prediction. Below is the detailed process for training and inference using this approach.

3.3.1 Training Process

The training process involves several key steps, starting with data preparation, followed by latent encoding, noise prediction, and backpropagation to refine the model. The following steps outline the full pipeline:

3.3.1.1 Data Preparation and Bounding Box Selection

Once we have extracted the bounding boxes for the driver's face or relevant regions (as discussed in the previous section), we proceed with centering the bounding box and creating a mask over the detected region.

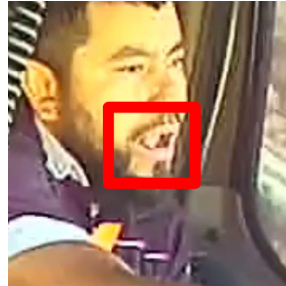


Figure 3.4: Center-cropped image based on bounding box for inpainting.

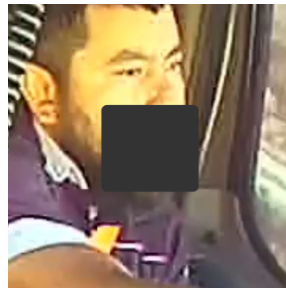


Figure 3.5: Masked image showing the region of interest for yawning inpainting.

3.3.1.2 Latent Encoding using VQ-VAE and Mask Creation

We utilize the VQ-VAE (Vector Quantized Variational Autoencoder) from **CompVis/latent-diffusion** to encode the input image into latent representations. A similar process is applied to the mask, where the bounding box is used to create the masked latent image.

- **Image Encoding:** The image is passed through the VQ-VAE model, which encodes it into a latent space representation.
- **Masked Latent Encoding:** The masked portion of the image is also encoded using VQ-VAE to create a masked latent representation.

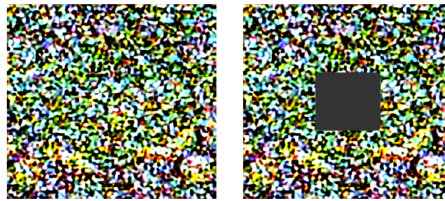


Figure 3.6: Latent encoding of the original and masked image.

3.3.1.3 Noising and Concatenation for Training

Once the latent image and masked latent image are generated, we apply noise to the latent image. The noisy latent image is then concatenated with the mask and masked latent image, creating a 7-channel input that will be passed through the U-Net.

- **Noising the Latent Image:** Gaussian noise is added to the latent image.
- **Concatenation:** The noisy latent image, mask, and masked latent image are concatenated to form the input tensor.



Figure 3.7: Concatenation of noisy latent, mask, and masked latent image.

3.3.1.4 Training the U-Net

The U-Net model is responsible for predicting the noise added to the latent image. The training objective is to minimize the difference between the actual noise added and the noise predicted by the U-Net. This is achieved through backpropagation, allowing the U-Net to improve its noise prediction capability over time.

- **Noise Prediction:** The U-Net predicts the noise added to the image.
- **Loss Function:** The difference between the actual and predicted noise is minimized using a standard loss function (e.g., MSE).
- **Backpropagation:** The model weights are updated based on the error signal from the loss function.

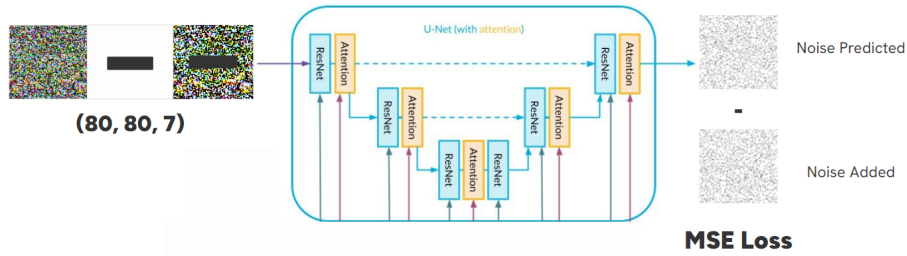


Figure 3.8: U-Net training to predict noise added to the latent image.

3.3.2 Inference Process

During inference, the trained model will be used to generate yawning images by denoising a random latent noise vector iteratively. The following steps outline the inference pipeline:

3.3.2.1 Latent Noise and Masked Image Input

For inference, we start by providing:

- Random latent noise
- A mask of the region where yawning will be synthesized (e.g., mouth region)
- A masked image of a normal person (e.g., with a closed mouth or normal expression)

These three inputs are concatenated to create the input for the U-Net.

3.3.2.2 Iterative Denoising Process

The U-Net is used iteratively to denoise the latent representation over a fixed number of steps, gradually turning the random noise into a realistic image of a driver who is yawning.

- **Denoising:** The U-Net progressively denoises the latent representation, refining the details in each step.
- **Fixed Steps:** The process is repeated for a pre-defined number of steps until a fully realistic image is generated.

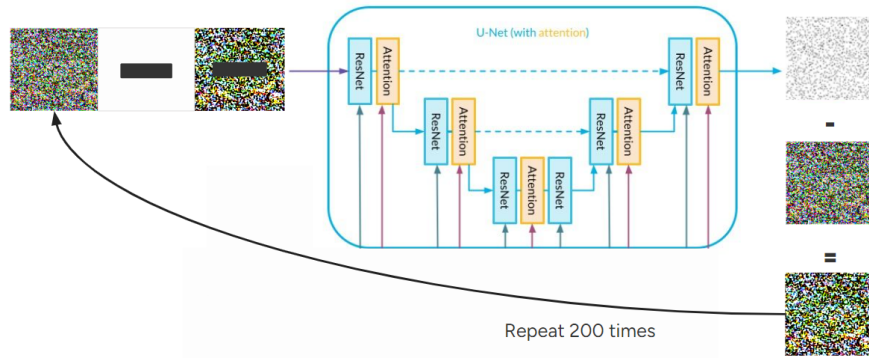


Figure 3.9: Iterative denoising process to generate yawning images.

3.3.2.3 Latent Decoding to Obtain Final Image

Once the latent representation has been fully denoised, it is decoded back into an image using the VQ-VAE decoder. The final image should depict the driver with a yawn.



Figure 3.10: Final yawning image decoded from the denoised latent representation.

This Dreambooth-style approach of inpainting with diffusion models allows us to generate synthetic images of drivers displaying underrepresented behaviors like yawning, which can then be used to augment the existing dataset and improve the performance of driver safety systems.

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